CO_2 and Photosynthesis

Pupil sheet

Name: _____ **Date:** _____

Safety: The box is not waterproof: never water the plant in the jar. Wash your hands after touching soil.

1. What I see on the screen

On the screen: **CO_2** (ppm), **Light** (lx), **Temperature** (°C), **Humidity** (%) and the **CO_2 rate**, a gradient with a ±. **If the lamp heats the jar, we have changed two factors at once.**

I read the gradient, not the value. After every change I wait **3 minutes**, then read the gradient over at least **3–5 minutes**. If the CO_2 falls towards **200 ppm** or the humidity passes **90%**, I air the jar for a minute and repeat the step.

In light I measure *net* photosynthesis (photosynthesis minus respiration). **Gross** = net + the respiration from darkness. A gradient differs from zero only if it is clearly larger than **twice its** ±.

2. What I do on my own

First I predict. In bright light the CO_2 will: ☐ ↑ ☐ ↓ ☐ → In darkness it will: ☐ ↑ ☐ ↓ ☐ →

Is there a light level at which the line stays flat? ☐ yes ☐ no

because _____

_____

The rig: plant _____ jar _____ L lamp at _____ cm

The basic experiments. Example: the app shows -10 ± 1 ppm/min over 5 min. Check by hand: 820 → 770 ppm in 5 min ⇒ $(770 - 820)/5 = -10$ ppm/min.

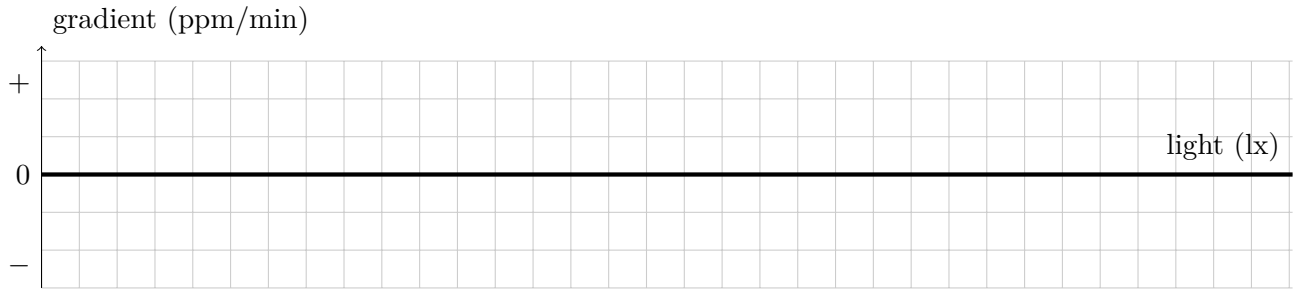
Condition	**Light (lx)**	**Minutes**	**Gradient** (ppm/min, with ±)
Empty jar (seal test)			
Pot without plant (soil)			
Plant, bright light			
Plant, darkness	0		

I subtract the soil's gradient. Net photosynthesis = _____ respiration = _____ gross = _____ ppm/min

The intensity series. At every step I check the temperature and press **Capture**.

Shading	**Light (lx)**	**Gradient** (ppm/min, with ±)	T **(°C)**
No sheet			
One sheet			
Two sheets			
Three sheets			
Darkness	0		

The graph. I plot the points. **The compensation point is where the curve crosses the thick zero line.**



Compensation point ≈ _____ lx (the app: _____ lx). Check: the value at 0 lx should be the respiration rate.

3. The conclusion I write

I predicted that, and the data show that _____

_____

When the line was flat, what was happening in the leaves was _____

_____

I cannot conclude anything about the planet's forests because _____

_____

Exit ticket

What is happening in the leaves when the CO_2 line is flat?

_____